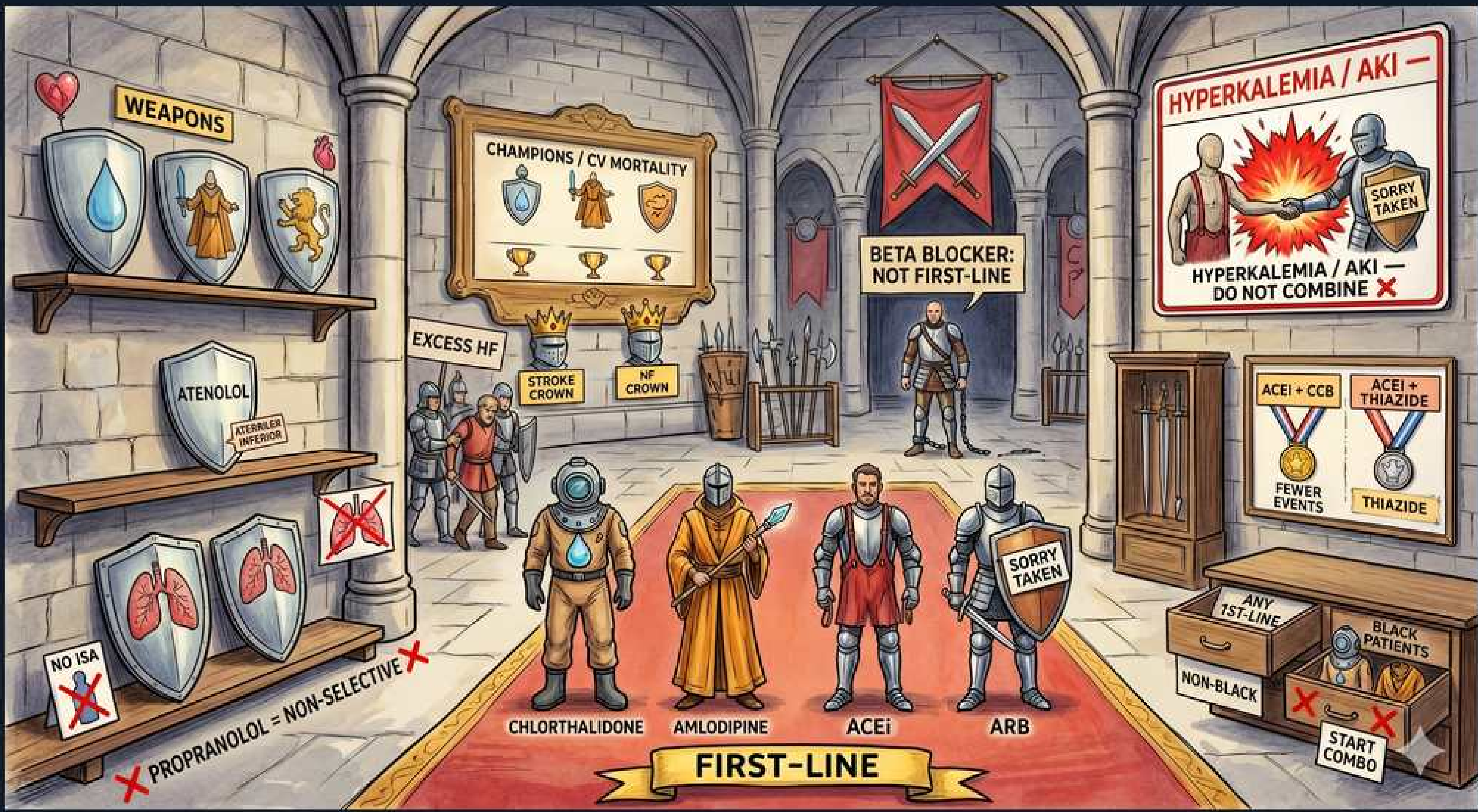
# Hypertension — First-Line Drug Classes



### 1. Atenolol shields (weapons rack)

**WHAT YOU SEE**

Shelf of shields/weapons with ATENOLOL and EXCESS HF labels.

**WHAT IT ENCODES**

Beta-blockers are NOT first-line for uncomplicated HTN — atenolol in particular underperforms for stroke prevention and excess HF/CV events versus other classes. Reserve beta-blockers for compelling indications: post-MI, HFrEF (carvedilol, metoprolol succinate, bisoprolol), atrial fibrillation rate control, and angina. Cardioselective (beta-1) agents (metoprolol, atenolol, bisoprolol) are preferred when a beta-blocker is needed in a patient with reactive airway disease.

**BOARD TRAP**

Selecting atenolol as initial monotherapy for a healthy 50-year-old with new Stage 2 HTN — WRONG. Beta-blockers are not first-line and atenolol is the worst-performing for stroke. The discriminator: only choose a beta-blocker for HTN when a compelling indication (post-MI, HFrEF, AF, angina) is present; otherwise pick a thiazide, ACEi/ARB, or CCB.

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### 2. No ISA / non-selective droplet

**WHAT YOU SEE**

Bottom shield: PROPRANOLOL = NON-SELECTIVE, NO ISA marked with X.

**WHAT IT ENCODES**

Non-selective beta-blockers (propranolol, nadolol) block beta-2 and can trigger bronchospasm — avoid in asthma/COPD; they also blunt and mask hypoglycemia symptoms in diabetics. Agents with intrinsic sympathomimetic activity (ISA: pindolol, acebutolol) are generally not preferred and are avoided post-MI. Non-selective beta-blockers without alpha activity can cause unopposed alpha vasoconstriction — dangerous in pheochromocytoma and cocaine-induced HTN unless alpha blockade is given first.

**BOARD TRAP**

Giving propranolol to an asthmatic hypertensive, or giving a pure beta-blocker first in suspected pheochromocytoma/cocaine toxicity — WRONG. The discriminator: non-selective beta-blockers cause bronchospasm and unopposed alpha vasoconstriction; alpha blockade (phentolamine) must precede beta blockade in catecholamine-excess states.

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### 3. Champions / CV mortality trophies

**WHAT YOU SEE**

A trophy/champions display case labelled CHAMPIONS / CV MORTALITY holding a STROKE CROWN and an HF CROWN — the trophies mark the four first-line classes as the proven winners at cutting stroke, HF and CV death (not just BP numbers).

**WHAT IT ENCODES**

The four first-line classes (thiazide-type diuretics, ACE inhibitors, ARBs, and dihydropyridine CCBs) are 'champions' because they reduce hard outcomes — stroke, heart failure, MI, and CV mortality — not merely BP numbers. Outcome trials (ALLHAT, ACCOMPLISH) underpin their preferred status. Choose among them based on comorbidities and demographics rather than treating all equally.

**BOARD TRAP**

Picking a drug solely because it lowers the BP number the most — WRONG. The discriminator: board questions reward agents proven to reduce cardiovascular events and mortality (the four first-line classes), not agents that merely produce the largest acute BP drop (e.g., hydralazine, clonidine).

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### 4. Beta-blocker: not first-line banner

**WHAT YOU SEE**

Red banner with knight: BETA BLOCKER NOT FIRST-LINE.

**WHAT IT ENCODES**

This reinforces that beta-blockers are excluded from first-line HTN therapy without a compelling indication, primarily because they are inferior for stroke prevention compared with ACEi/ARB, CCB, and thiazides. They also do not regress LVH as well and have less favorable metabolic profiles. Their place is comorbidity-driven (post-MI, HFrEF, AF, angina).

**BOARD TRAP**

Adding a beta-blocker as the second agent for a patient already on a thiazide simply to intensify therapy — WRONG when no compelling indication exists. The discriminator: prefer logical combinations (ACEi/ARB + CCB, or ACEi/ARB + thiazide); a beta-blocker is added only for a specific cardiac indication.

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### 5. Hyperkalemia / AKI warning

**WHAT YOU SEE**

Top-right poster: HYPERKALEMIA / AKI, DO NOT COMBINE (ACEI + ARB).

**WHAT IT ENCODES**

NEVER combine an ACE inhibitor with an ARB (or with a direct renin inhibitor like aliskiren): dual RAAS blockade causes hyperkalemia and AKI with NO added cardiovascular benefit (ONTARGET, ALTITUDE trials). Monitor K+ and creatinine 1-2 weeks after starting or up-titrating any single RAAS agent; a creatinine rise up to ~30% is acceptable and expected, but a larger jump suggests bilateral renal artery stenosis.

**BOARD TRAP**

Adding an ARB to a patient already on an ACE inhibitor to improve proteinuria or BP — WRONG. This is a classic distractor. The discriminator: ACEi + ARB increases hyperkalemia and AKI without outcome benefit and is contraindicated; switch agents rather than stack them.

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### 6. Chlorthalidone (knight 1)

**WHAT YOU SEE**

First of four armored 'FIRST-LINE' knights standing on the red carpet, shield labelled CHLORTHALIDONE — a knight in the champions' lineup = the preferred thiazide diuretic.

**WHAT IT ENCODES**

Chlorthalidone is a thiazide-TYPE diuretic and the preferred thiazide because of its long half-life (~45-60h, 24-hour control) and proven outcome data (ALLHAT, SHEP). Typical dose 12.5-25 mg daily. Thiazides inhibit the Na-Cl cotransporter in the distal convoluted tubule. They are especially effective in Black patients, the elderly, and salt-sensitive HTN. Monitor for hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, and hypercalcemia.

**BOARD TRAP**

Treating chlorthalidone and hydrochlorothiazide as interchangeable at equal doses — WRONG. The discriminator: chlorthalidone is roughly 1.5-2x more potent, much longer-acting, and has stronger CV outcome evidence; when outcomes matter, chlorthalidone is the better answer.

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### 7. Amlodipine (knight 2)

**WHAT YOU SEE**

Second first-line knight in the lineup, shield labelled AMLODIPINE — the dihydropyridine CCB champion.

**WHAT IT ENCODES**

Amlodipine is a dihydropyridine CCB and an excellent first-line agent, particularly effective in Black patients and the elderly. Dose 5-10 mg daily. DHP CCBs cause peripheral vasodilation; the main side effect is dose-dependent peripheral edema (lower-extremity), which is reduced by adding an ACEi/ARB rather than a diuretic. Unlike non-DHP CCBs (verapamil, diltiazem), amlodipine does not depress AV conduction or contractility, so it is safe in HFrEF.

**BOARD TRAP**

Treating amlodipine-induced ankle edema by adding furosemide — WRONG. The edema is from precapillary vasodilation, not fluid overload, so a diuretic does not help. The discriminator: add an ACEi/ARB (balances post-capillary tone) or reduce the dose; also do not confuse safe amlodipine with verapamil/diltiazem, which are contraindicated in HFrEF.

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### 8. ACEI (knight 3)

**WHAT YOU SEE**

Third first-line knight, shield labelled ACEI — the ACE-inhibitor champion in the row of four.

**WHAT IT ENCODES**

ACE inhibitors (-pril) block conversion of angiotensin I to II, lowering BP and reducing aldosterone. First-line and especially preferred in diabetes or CKD with albuminuria (renoprotective via efferent arteriolar dilation), HFrEF, and post-MI. Side effects: dry cough (~10%, from bradykinin), hyperkalemia, acute creatinine rise, and angioedema (more common in Black patients; life-threatening). Contraindicated in pregnancy and bilateral renal artery stenosis.

**BOARD TRAP**

Switching an ACEi to a different ACEi because of cough, or continuing an ACEi after angioedema — WRONG. The discriminator: ACEi cough is a class effect, so switch to an ARB; ACEi-induced angioedema is an absolute contraindication to all ACEi and warrants caution even with ARBs.

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### 9. ARB (knight 4)

**WHAT YOU SEE**

Fourth first-line knight with a SORRY TAKEN tag, shield labelled ARB — the ARB champion, 'taken' because it is never paired with the ACEi knight beside it.

**WHAT IT ENCODES**

ARBs (-sartan) block the angiotensin II type-1 receptor, giving the same first-line benefits as ACEi (diabetes/CKD with albuminuria, HFrEF, post-MI) without bradykinin accumulation — so they cause far LESS cough and much less angioedema. Like ACEi, they cause hyperkalemia and creatinine rise, are contraindicated in pregnancy and bilateral renal artery stenosis, and must never be combined with an ACEi.

**BOARD TRAP**

Assuming an ARB is completely safe after ACEi angioedema — partially WRONG. Cough resolves with an ARB, but angioedema can rarely still occur, so use cautiously. The discriminator: ARB is the right substitute for ACEi cough; for ACEi angioedema it is used only with caution, never as a casual swap.

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### 10. ACEI + CCB combo

**WHAT YOU SEE**

A wooden crate stamped ACEI + CCB, FEWER EVENTS — the boxed pairing that ACCOMPLISH proved beats ACEi+thiazide on hard outcomes.

**WHAT IT ENCODES**

ACEi + dihydropyridine CCB is the PREFERRED two-drug combination based on ACCOMPLISH, which showed fewer CV events than ACEi + thiazide (benazepril/amlodipine beat benazepril/HCTZ). This pairing is mechanistically complementary, metabolically neutral, and the CCB-induced edema is offset by the ACEi. Combination therapy is recommended initially for Stage 2 HTN or BP >20/10 above goal.

**BOARD TRAP**

Defaulting to ACEi + thiazide for every patient needing two drugs — not WRONG, but the ACEi + CCB combination has superior outcome data (ACCOMPLISH). The discriminator: when the question pits the two combos against each other on hard endpoints, ACEi + CCB is the better-supported answer.

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### 11. ACEI + thiazide combo

**WHAT YOU SEE**

A wooden crate stamped ACEI + THIAZIDE — the acceptable but second-best two-drug combo box.

**WHAT IT ENCODES**

ACEi + thiazide is an acceptable and effective combination — the thiazide-induced RAAS activation and mild hypokalemia are counterbalanced by the ACEi's RAAS blockade and K+-retaining effect. It is widely used and available as fixed-dose combinations. It controls volume and is reasonable, but ACCOMPLISH showed ACEi + CCB reduced events more.

**BOARD TRAP**

Combining an ACEi with a potassium-sparing diuretic and a K+ supplement together — WRONG (hyperkalemia). The discriminator: ACEi + thiazide is safe and K-balanced, but stacking multiple K-retaining strategies on top of an ACEi causes dangerous hyperkalemia.

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### 12. Any first-line / non-Black

**WHAT YOU SEE**

A box reading ANY 1st LINE for NON-BLACK patients — any of the four champion classes works as starter monotherapy here.

**WHAT IT ENCODES**

In non-Black adults without a compelling indication, any of the four first-line classes (thiazide, ACEi, ARB, CCB) is appropriate initial monotherapy, with selection guided by comorbidities (e.g., ACEi/ARB if diabetes/CKD/HFrEF). For Stage 2 or BP >20/10 above goal, start two first-line agents from different classes.

**BOARD TRAP**

Believing one specific class is mandatory first-line for all uncomplicated non-Black patients — WRONG. The discriminator: in the absence of a compelling indication, the four classes are interchangeable as initial therapy; comorbidities, not a fixed hierarchy, drive the choice.

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### 13. Black patients -> CCB/thiazide

**WHAT YOU SEE**

A box reading BLACK PATIENTS — START COMBO — pointing toward CCB/thiazide initial therapy because low-renin physiology blunts ACEi/ARB monotherapy.

**WHAT IT ENCODES**

In Black adults WITHOUT HF or CKD, initial therapy should be a CCB or thiazide (often as initial combination), because lower-renin physiology makes ACEi/ARB monotherapy less effective and ACEi angioedema is more common. The exception: if the Black patient has CKD with albuminuria or HFrEF, an ACEi/ARB IS indicated for its compelling renal/cardiac benefit.

**BOARD TRAP**

Starting ACEi/ARB monotherapy in a Black patient with uncomplicated HTN — WRONG (less effective, higher angioedema risk). But equally WRONG is withholding an ACEi/ARB from a Black patient who has diabetic CKD with proteinuria. The discriminator: race-based preference for CCB/thiazide applies only without a compelling indication; comorbidity always overrides it.